

# Fifty Tokens of Signal: A Cross-Dataset Audit of Overlap, Shortcuts, and Label Divergence in Online-Abuse Benchmarks

Dhruv Patel

Independent Researcher

dhruv17062000@gmail.com

## Abstract

A decade after the first Workshop on Online Abuse and Harms, abuse detection is evaluated on a sprawling collection of public datasets that are usually treated as independent and interchangeable. We audit five widely used English benchmarks (Davidson, DynaHate, HateXplain, TweetEval-Hate, and the Stormfront corpus; 109,698 posts) and, by measuring contamination, lexical shortcuts, label conflict, and transfer on one common scale, argue a single thesis: for these benchmarks the evidence points to a conceptual measurement bottleneck (what counts as abuse) more than to data hygiene (contamination or leakage), a reading we hold with the caution that the strict-cutoff conflict sample is small. Three findings carry this claim. (i) The data-hygiene worries are small here: cross-dataset duplication is negligible (only 17 near-duplicate pairs at Jaccard  $\geq 0.8$  across 108,946 unique texts, 5 of them exact), and train/test near-duplicate leakage, though common (up to 9.8% of test items), barely changes macro-F1 for the pipeline we study, both for TF-IDF logistic regression and for a fine-tuned distilBERT on the two official-split corpora. (ii) The signal these benchmarks reward is concentrated in a few tokens: a TF-IDF model restricted to 50 tokens retains roughly 89 to 95% of full-model macro-F1 on every dataset, and, against the natural objection that this is a bag-of-words artifact, a fine-tuned distilBERT trained on only those 50 tokens per post still retains 81 to 92% across all five datasets (lowest on the adversarially debiased DynaHate). (iii) Where the same surface text recurs across corpora, its binary labels frequently flip, isolating definition from sampling: at a loose Jaccard  $\geq 0.5$  near-duplicate cutoff the clean per-item conflict rate is 48.6% (103 of 212 shared texts, though many of these loose matches share a profanity skeleton rather than identical text); at the strict same-text Jaccard  $\geq 0.8$  cutoff the clean shared set is small ( $n = 18$ , 10 in conflict) and the disagreement is a Davidson-driven

offensive-versus-hate split. Transfer is correspondingly poor (a 0.20 macro-F1 gap) but, given negligible contamination, reflects definitional and distributional divergence rather than leakage; we cannot separate those two, so the conceptual claim rests on the item-level comparison rather than on the transfer gap. Holding the near-duplicate graph fixed and sending both corpora’s *offensive* class to not-abuse moves the strict-cutoff conflict rate from 55.6% to 22.2%, so much of that disagreement is the offensive-versus-hate boundary rather than disagreement about hate itself. We release *abuse-audit*, a one-command tool that emits the same per-dataset report card for any new corpus, and recommend reporting a shortcut score and a cross-dataset transfer cell as standard practice.

**Content warning.** This paper quotes masked slurs and abusive posts from public datasets because they are the object of the audit; examples are kept to the minimum the argument needs.

## 1 Introduction

The Workshop on Online Abuse and Harms turns ten this year. Over that decade the community produced dozens of labeled datasets for hate speech, toxicity, and abuse, and a now-standard practice grew around them: train on one corpus, report a macro-F1, and sometimes claim generalization by testing on another. Three assumptions underpin this practice. The datasets are treated as largely independent, so combining or comparing them is taken to be sound. A high score is read as evidence that a model has learned something about abuse. And when scores fail to transfer, the cause is waved off as distributional.

This paper audits those assumptions directly. We take five of the most widely used public English benchmarks, normalize them to a common schema, and measure three things with a single reproducible pipeline: how much the datasets overlap (and whether overlapping items agree on labels),

how much of a model’s performance is recoverable from a handful of tokens, and how well models transfer across datasets. We deliberately use simple, transparent, classical methods (character n-gram MinHash for near-duplicate detection, TF-IDF with logistic regression for classification) so that every number can be reproduced on a laptop with no GPU and no paid API, and so that the findings reflect properties of the data rather than of a particular large model.

Our contribution is a single thesis with one new piece of method behind it: for these benchmarks the evidence points to a conceptual measurement bottleneck more than to data hygiene (a reading held with the caution that the strict-cutoff conflict sample has only 18 items), and, to our knowledge, no prior work puts all four axes (contamination, leakage, lexical shortcut, and item-level label conflict) on one common, reproducible scale, weighed against each other rather than studied one corpus at a time. Three results carry the thesis, and we lead with the two that are best powered. First, the data-hygiene problems the community most often polices are minor here: cross-corpus near-duplicate contamination is negligible (so poor transfer is not leakage between corpora), and train/test near-duplicate leakage, though common, barely changes macro-F1 for the pipeline we study, in both the linear model and a fine-tuned distilBERT on the two official-split corpora. Second, the signal these benchmarks reward is concentrated in a handful of tokens: with one comparable shortcut score computed identically across five corpora we show that a few tokens recover most of the measured signal for the standard linear pipeline, and, addressing the obvious bag-of-words objection, a small contextual model (distilBERT) trained on only those tokens retains almost all of its full-text performance, so the concentration is a property of the labeled data rather than of TF-IDF. Third, as supporting and illustrative evidence we contribute an item-level cross-corpus label-conflict measurement over these benchmarks, not previously reported to our knowledge: where the same surface text recurs across datasets its labels frequently flip, which isolates definition from annotator noise and sampling in a way guideline-level comparisons cannot. We are explicit about its power: the near-50% rate holds at a loose Jaccard  $\geq 0.5$  cutoff (where many matches share a profanity skeleton rather than identical text), while at the strict  $\geq 0.8$  cutoff the clean sample is small and Davidson-driven,

so we treat it as a window onto definitional divergence rather than the headline number. To make the audit reusable we release `abuse-audit`, a one-command open-source tool that emits this report card for any abuse dataset CSV and reproduces every per-dataset number here on a laptop CPU. Fitting for a tenth-anniversary workshop, we take these findings as a case for treating the definition of the task itself, not data hygiene alone, as the main measurement bottleneck.

## 2 Related Work

**Leakage and overfitting.** Arango et al. (2020) showed that reported hate-speech results collapse once two flaws are fixed: feature extraction over the combined train and test sets, and author-level leakage, with a large fraction of one popular corpus authored by a handful of users who appear in both splits. Geva et al. (2019) generalize the concern to annotator fingerprints. These studies establish within-dataset leakage; we test whether near-duplicate leakage matters for standard linear pipelines and whether it extends across corpora.

**Shortcuts and bias.** A line of work documents that abuse classifiers latch onto surface cues. Wiegand et al. (2019) trace this to biased sampling, Dixon et al. (2018) to identity-term imbalance, Kennedy et al. (2020) to over-attention to group identifiers, and Ramponi and Tonelli (2022) disentangle spurious from authentic lexical artifacts on single datasets. Gröndahl et al. (2018) show models transfer poorly and are easily evaded. These works establish lexical bias largely qualitatively or on a single corpus; our delta is one comparable retained-F1 shortcut score computed identically across five corpora (and a sixth, implicit-hate, stress test), so the datasets can be ranked on a common scale. We do not propose a new bias metric.

**Cross-dataset generalization and definitions.** Fortuna et al. (2020) show that label definitions are inconsistent across datasets, and Fortuna et al. (2021) and Antypas and Camacho-Collados (2023) measure transfer failure and tie it to label-space mismatch. Concurrent WOA work by Melis et al. (2025) decomposes hate-speech definitions into modular criteria and shows that a classifier’s zero-shot performance shifts with which criteria a prompt encodes; that result and ours are complementary evidence for the same conclusion, that the definition used to label or prompt a system, not

only the data or the model, is doing much of the work, though their probe is on prompted LLMs and ours is on the labels already fixed in five released benchmarks. More recently, [Tonneau et al. \(2025\)](#) evaluate models on a sample representative of a day on Twitter and find that academic benchmarks greatly overestimate real-world performance, attributing much of the gap to models confusing hate with merely offensive speech, the same definitional fault line our cross-corpus conflicts fall along. [Davidson et al. \(2019\)](#) and [Sap et al. \(2019\)](#) document demographic label bias, and a related line measures disagreement among annotators labeling the same posts within one dataset. These studies establish the divergence at the level of guidelines, label spaces, aggregate performance, or annotators of a shared corpus. From any of these one can infer that definitions differ, but not rule out that the apparent disagreement is annotator noise or sampling, since the texts being compared are not the same texts. Our delta is a sharper, falsifiable claim those designs could not make: holding the surface text fixed, the *same* post flips binary label across independently published corpora, isolating the corpus’s definition from annotator disagreement within a corpus and from differences in what each corpus sampled. We are equally explicit about what this does not buy: the shared items are too few to mechanically explain the transfer gap (§4.4), so we offer the item-level conflict as direct evidence that the fault line is definitional, not as the quantitative cause of poor transfer.

**Catalogs, diagnostics, and merging.** [Vidgen and Derczynski \(2020\)](#) catalog dozens of datasets; [Röttger et al. \(2021\)](#) introduce functional tests for models. Recent efforts merge many corpora, for example MetaHate ([Piot et al., 2024](#)), the cross-taxonomy integration of [Fillies and Paschke \(2025\)](#), and the composition analysis of [Guimarães et al. \(2023\)](#), but deduplicate only within each source and reconcile labels at the level of taxonomies rather than checking the shared items. We are not aware of prior work that measures near-duplicate overlap across these corpora and releases a reusable per-dataset audit. We position `abuse-audit` as a data-side analogue of HateCheck.

### 3 Data and Method

**Datasets.** We use five public, no-login English datasets, summarized in Table 1. They span sources (Twitter, the Gab and Stormfront platforms, and

| Dataset                                               | Source      | <i>N</i> | % abuse |
|-------------------------------------------------------|-------------|----------|---------|
| Davidson ( <a href="#">Davidson et al., 2017</a> )    | Twitter     | 24,783   | 83.2    |
| DynaHate ( <a href="#">Vidgen et al., 2021</a> )      | Synthetic   | 41,144   | 53.9    |
| HateXplain ( <a href="#">Mathew et al., 2021</a> )    | Twitter/Gab | 20,148   | 59.2    |
| TweetEval ( <a href="#">Barbieri et al., 2020</a> )   | Twitter     | 12,962   | 42.1    |
| Stormfront ( <a href="#">de Gibert et al., 2018</a> ) | Forum       | 10,661   | 11.2    |
| Total                                                 |             | 109,698  |         |

Table 1: The five audited benchmarks. All are publicly downloadable without login, API key, or tweet rehydration.

synthetic adversarial generation) and label schemes. We map each to a binary label (abuse or hate positive vs. not) using documented per-dataset rules, and retain the native labels for reference. Texts are normalized by lowercasing, removing URLs, replacing user mentions with a placeholder, stripping retweet markers, and collapsing whitespace. After dropping empty rows we have 109,698 posts and 108,946 unique normalized texts.

**Overlap detection.** Figure 1 summarizes the whole pipeline. We collapse the corpus to unique (dataset, normalized text) nodes. We compute character 5-gram shingles per node and estimate Jaccard similarity with MinHash (128 permutations) indexed by locality-sensitive hashing ([Broder, 1997](#)). Candidate pairs returned by LSH are verified with exact shingle Jaccard. We report pair counts at thresholds 0.5, 0.7, 0.8, and 0.9 for sensitivity (Appendix E), and treat exact duplicates separately. A pair is cross-dataset if its two nodes come from different datasets.

**Shortcut score.** For each dataset we split 80/20 (stratified) and train a TF-IDF (unigram and bigram) logistic-regression model as the full reference. We then select the 50 unigrams with the highest  $\chi^2$  association with the label on the training split and train a model restricted to those 50 tokens. The shortcut score is the ratio of top-50 macro-F1 to full macro-F1. A high ratio means a handful of tokens carries most of the signal. We report bootstrap 95% confidence intervals over the test set (1,000 resamples). Two caveats fix the interpretation. The score measures how concentrated the *linear-recoverable* signal is for this TF-IDF pipeline, not how lexical the task is in some absolute sense: a stronger encoder could extract structure that unigrams cannot. The comparison is also not perfectly clean, because the full reference uses unigrams and bigrams while the top-50 model uses unigrams only, so the retained fraction folds

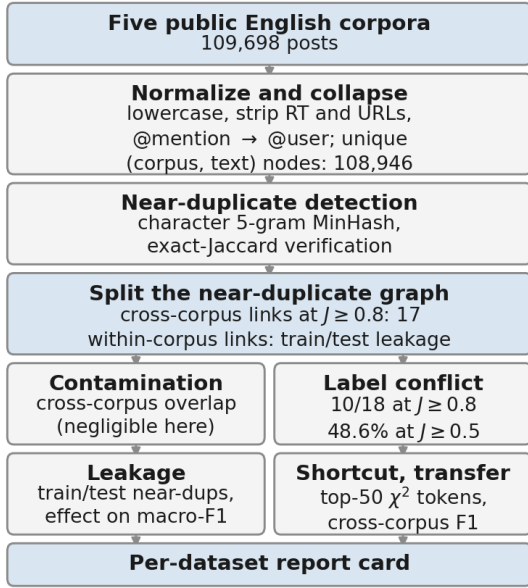


Figure 1: The audit pipeline. One pass over the five corpora produces the near-duplicate graph, and every measurement in Section 4 is read off that graph or off the same normalized nodes, so the four axes are computed on one common basis rather than assembled from separately tuned analyses. Counts are generated from the released artifacts by `code/make_pipeline_figure.py`.

together the effect of keeping fewer features and the effect of dropping all bigrams. We read a high score as evidence that the signal these models use is concentrated in a few tokens, and we revisit the asymmetry in Section 4.3.

**Transfer.** We train one TF-IDF logistic-regression model per dataset and evaluate macro-F1 on every dataset. Off-diagonal cells train on all of dataset A and test on all of dataset B. Diagonal cells use a held-out 80/20 split to avoid train-on-test inflation.

**Leakage.** For datasets with official splits (DynaHate, TweetEval) we use them; for the others we average over three random 80/20 splits. We flag each test item that exactly or near-duplicates (Jaccard  $\geq 0.8$ ) any training item, then compare macro-F1 on the full test set against the test set with leaked items removed.

## 4 Results

### 4.1 Cross-dataset duplication is negligible

Across 108,946 unique texts, only 17 cross-dataset near-duplicate pairs at Jaccard  $\geq 0.8$  span datasets, of which 5 are exact duplicates (Appendix B, Figure 4). The overlap is dominated by, but not confined to, the Twitter-sourced corpora: at Jaccard  $\geq 0.8$  the synthetic DynaHate still participates in 6 of the 17 pairs (and in 3 of the 5 exact duplicates), mostly short, generic strings such as “*i hate you*”. The Stormfront forum corpus is the genuinely disjoint case, with no cross-dataset pairs at this threshold and only 15 (of 272) at the loose Jaccard  $\geq 0.5$  cutoff. Within datasets, by contrast, near-duplication is common (5,201 near-duplicate node pairs at the same threshold). Merging these corpora therefore does not silently double count, but the transfer failures reported below cannot be blamed on contamination either.

### 4.2 Where texts overlap, labels disagree

We measure the conflict rate at the level of *items* (distinct shared texts) rather than pairs. A node is in conflict if any of its cross-dataset near-duplicate neighbors carries a different binary label; the denominator is the distinct nodes that have at least one such neighbor. This avoids the bias of a pair-weighted count, in which a text that recurs in several corpora contributes once per pair (95 of the 281 nodes at Jaccard  $\geq 0.5$  appear in more than one pair, one in 16), and it makes a node-level interval the natural uncertainty estimate. We also drop placeholder and normalization artifacts, namely nodes that are at least half user-mention tokens or have fewer than fifteen non-placeholder characters (strings such as @user @user or “*i hate you*”), so the rate is not carried by near-duplicate matches that are essentially the mention placeholder.

Our primary figure is the clean (artifact-removed) item-level rate at the Jaccard  $\geq 0.5$  cutoff, the largest statistically usable sample: 103 of 212 such nodes are in conflict, 48.6% (95% bootstrap-over-nodes CI [42.0, 55.7], 10,000 resamples). The estimate is stable as the cutoff is tightened to genuine near-duplicates: at Jaccard  $\geq 0.8$  the clean item-level rate is 55.6% (10 of 18; CI [33.3, 77.8]), where the interval is wide because clean high-similarity overlap is itself scarce ( $n = 18$ ); the all-nodes rate at Jaccard  $\geq 0.5$  is 53.7% (151 of 281; CI [48.0, 59.4]). One caveat on the interval: the two endpoints of each cross-dataset



near-duplicate are correlated nodes, so a bootstrap over nodes is a lower bound on the true uncertainty. Collapsing each connected component of the near-duplicate graph to one independent unit (a cluster of mutually near-duplicate texts, counted as conflicting if its members span both labels) gives 31 of 73 clean components in conflict at Jaccard  $\geq 0.5$ , 42.5% (CI [31.5, 53.4]), and 4 of 8 at Jaccard  $\geq 0.8$ , 50.0% (CI [12.5, 87.5]). The point estimate slides down and the interval widens, but the node-level 48.6% still sits inside the component interval, so the “about half” reading survives at this loose cutoff. We are careful about what “same text” means here: at Jaccard  $\geq 0.5$  many of the matched pairs are not the same content but share a profanity skeleton (for example “*happy birthday b\*\*\*h*” versus “*happy 4th birthday big b\*\*\*h*”), so the 48.6% is the loose near-duplicate rate rather than a rate over identical text. At the genuinely-near-duplicate Jaccard  $\geq 0.8$  cutoff the shared items are scarce ( $n = 18$ ) and the conflicting cross-dataset links are dominated by Davidson’s offensive-versus-hate distinction (7 of the 9 such links involve Davidson; Table 8), so the strict-cutoff disagreement is a small-sample, definitional, offensive-versus-hate phenomenon. This is the item-level form of the definitional divergence that Fortuna et al. (2020) identified at the guideline level and that Tonneau et al. (2025) infer from real-world performance gaps. The conflicts are interpretable: most arise because the broad “offensive language” class in Davidson marks generic profanity as positive while the hate-specific corpora do not. For instance “*karma is a b\*\*\*h*” is positive in Davidson but negative in both HateXplain and DynaHate, and “*you p\*\*\*y*” is positive in Davidson but negative in TweetEval (matches are near-duplicates, so paired texts can differ by punctuation or a user mention; a subset are byte-identical). Table 7 in Appendix A lists every conflicting link at the strict cutoff with our reading of each. It also shows how narrow that evidence is: the nine links reduce to six distinct texts, and one of them, “*karma is a b\*\*\*h*”, accounts for four. Two of the six are not the offensive-versus-hate split at all. “*i f\*\*\*\*\*g hate myself*” is self-directed and has no target group, and one is reclaimed in-group usage, so the binary schemes are also splitting on target and on speaker stance, not only on severity.

The finding does not rest on Davidson or on a few high-degree nodes. Removing Davidson, the main source of conflicts, leaves a pair-weighted rate of 47.6% (49 of 103 cross-dataset pairs at Jaccard

$\geq 0.5$ ), essentially unchanged from the 47.8% over all pairs. The item-level Davidson-excluded rate depends on the cutoff: at Jaccard  $\geq 0.5$  it is if anything higher (58.7%,  $n = 109$  nodes), while at the stricter Jaccard  $\geq 0.8$  it falls to 40.0% (4 of 10), a drop that is noisy at that small  $n$ ; the stable robustness statement is therefore at Jaccard  $\geq 0.5$ . Dropping the highest-degree nodes (the texts that recur most widely) also barely moves the item-level rate at Jaccard  $\geq 0.5$ : it stays between 48.7% and 49.8% as the top one, three, five, or ten nodes are removed (Table 3). The same or near-identical string is abuse or not depending mostly on which corpus it was drawn from.

**What the binary collapse costs.** Two of the five corpora fold generic offensive language in with hate to form our positive class (Davidson: 1,430 *hate* plus 19,190 *offensive*; HateXplain: 5,935 *hatespeech* plus 6,399 *offensive*). To measure what that collapse destroys we hold the near-duplicate graph fixed, since it is computed from text alone and does not depend on labels, and vary only the mapping (Table 2, from `code/label_mapping_sensitivity.py`, which refuses to report a variant unless it first reproduces the published baseline row exactly). At the strict cutoff the collapse accounts for much of the disagreement but not all of it: sending Davidson’s *offensive* class to the negative side moves the clean rate from 55.6% to 44.4%, and sending both corpora’s *offensive* classes moves it to 22.2% (4 of 18). We do not over-read the size of that drop, because at  $n = 18$  the intervals overlap ([33.3, 77.8] against [5.6, 44.4]). At the loose cutoff the effect is not monotone in how conservative the mapping is, which is the more informative result: remapping Davidson alone raises the rate to 55.7%, and remapping both lowers it only to 46.7% from 48.6%. The reason is that a stricter mapping relocates disagreement rather than removing it. Going from the published mapping to the Davidson-only mapping, 42 nodes stop conflicting while 57 begin to, and 27 of those 57 are nodes whose own label never changed and whose neighbour’s did. Choosing a more conservative binary scheme therefore does not make cross-corpus disagreement go away.

**Robustness to a sixth corpus.** The strict-cutoff sample ( $n = 18$  clean items) is the weakest point of the conflict finding, so we ran the identical overlap and item-level pipeline

| Binary mapping                                                   | $J \geq 0.8$ |                   | $J \geq 0.5$ |                   |
|------------------------------------------------------------------|--------------|-------------------|--------------|-------------------|
|                                                                  | $n$          | Rate %            | $n$          | Rate %            |
| Published mapping (hate or offensive = abuse)                    | 10/18        | 55.6 [33.3, 77.8] | 103/212      | 48.6 [42.0, 55.7] |
| Davidson <i>offensive</i> $\rightarrow$ not abuse                | 8/18         | 44.4 [22.2, 66.7] | 118/212      | 55.7 [49.1, 62.3] |
| Davidson and HateXplain <i>offensive</i> $\rightarrow$ not abuse | 4/18         | 22.2 [5.6, 44.4]  | 99/212       | 46.7 [40.1, 53.3] |

Table 2: Sensitivity of the clean item-level cross-dataset conflict rate to the binary label mapping, with the near-duplicate graph held fixed so only the mapping varies. Bootstrap 95% intervals in brackets. Generated by `code/label_mapping_sensitivity.py`, which refuses to run unless it first reproduces the published baseline row exactly.

(`code/expanded_conflict.py`, with new artifacts and appended checksums; the frozen five-dataset numbers are untouched) over the five benchmarks plus a sixth public, no-login corpus, OLID/OffensEval-2019 (Zampieri et al., 2019), mapping its subtask-A OFF/NOT labels to our binary scheme. We had intended to add HatEval (Basile et al., 2019) as a seventh; we could not, because every public redistribution of HatEval omits the tweet text to comply with the original terms (the gold test labels ship in a password-encrypted archive and the full text is behind a registration form), and our text-based overlap pipeline cannot run on labels alone, so we stop on that corpus rather than substitute the already-included TweetEval repackaging. Adding OLID enlarges the cross-dataset overlap: the strict Jaccard  $\geq 0.8$  clean item sample grows from 18 to 22 and the cross-dataset pair count from 17 to 29, while the loose Jaccard  $\geq 0.5$  clean sample grows from 212 to 248. The conclusions do not change. At the loose cutoff the clean item-level conflict rate is 49.2% (122 of 248; CI [43.1, 55.2]), essentially the frozen 48.6%, so the “about half” reading holds on the larger sample. At the strict cutoff the rate is 45.5% (10 of 22; CI [22.7, 63.6]), still about half and still small- $n$ . Crucially, all twelve of OLID’s cross-dataset near-duplicate links at Jaccard  $\geq 0.8$  agree on the label (its offensive class aligns with Davidson’s offensive class and with TweetEval), so OLID adds shared items but *no* new conflicting links: the nine conflicting links are the same as before, and Davidson still participates in seven of them. The expanded result therefore enlarges the sample without broadening the phenomenon: the strict-cutoff disagreement remains the same small-sample, Davidson-driven, offensive-versus-hate split rather than a pattern that generalizes to a new corpus.

| Conflict-rate estimate                                                                   | Jacc. $\geq 0.5$          | Jacc. $\geq 0.8$  |
|------------------------------------------------------------------------------------------|---------------------------|-------------------|
| <i>Item-level (per distinct shared text), <math>n</math> in parentheses</i>              |                           |                   |
| Clean (primary)                                                                          | 48.6 ( $n_{212}$ )        | 55.6 ( $n_{18}$ ) |
| bootstrap-over-nodes CI                                                                  | [42.0, 55.7]              | [33.3, 77.8]      |
| All nodes                                                                                | 53.7 ( $n_{281}$ )        | 51.9 ( $n_{27}$ ) |
| Clean, Davidson excl.                                                                    | 58.7 ( $n_{109}$ )        | 40.0 ( $n_{10}$ ) |
| <i>Component-level (one unit per shared-text cluster)</i>                                |                           |                   |
| Clean                                                                                    | 42.5 ( $n_{73}$ )         | 50.0 ( $n_8$ )    |
| bootstrap CI                                                                             | [31.5, 53.4]              | [12.5, 87.5]      |
| <i>Pair-weighted (each cross pair counts once)</i>                                       |                           |                   |
| All pairs                                                                                | 47.8                      | 52.9              |
| Davidson excl.                                                                           | 47.6                      | 40.0              |
| <i>Item-level, all nodes, top-<math>k</math> nodes removed (<math>J \geq 0.5</math>)</i> |                           |                   |
| $k = 1 / 3 / 5 / 10$                                                                     | 48.9 / 51.0 / 48.7 / 49.8 |                   |

Table 3: Cross-dataset label-conflict rate (%) under item-level, component-level, and pair-weighted counting, with artifact filtering, Davidson exclusion, and removal of the highest-degree nodes. The primary figure is the clean item-level rate at Jaccard  $\geq 0.5$ ; it is about one half and stable across these choices. The clean rate exceeds the all-nodes rate at the strict cutoff because the dropped artifact rows (mention placeholders, very short strings) are disproportionately non-conflicting. Because the two endpoints of each near-duplicate are correlated, the bootstrap-over-nodes interval is a lower bound on the true uncertainty; the component-level block treats each shared-text cluster as one independent unit, which widens the interval and lowers the point estimate while leaving the “about half” reading intact.

### 4.3 Signal is concentrated in a few tokens for the linear pipeline

Table 4 and Figure 2 report the shortcut score. On every dataset, a model restricted to 50 tokens retains between 89.2% and 95.4% of the full model’s macro-F1 (intervals are tight except on the small Stormfront set). The choice of 50 is not load-bearing: a budget sweep (Appendix F) shows retention rising from 63 to 93% at 10 tokens to 89 to 95% at 50 and 92 to 100% at 100, and even the 50-token model sits far above a majority-class baseline (macro-F1 0.35 to 0.47, versus 0.61 to 0.82 for the 50-token model). The highest- $\chi^2$  tokens are

dominated by slurs, identity terms, and a small set of keywords (Table 5). Under this measurement, abuse detection on these benchmarks is close to keyword spotting for a linear model: a few tokens carry most of the signal that TF-IDF logistic regression can exploit. The headline retained percentage compares the top-50 unigram model against the unigram-plus-bigram full reference used throughout, so on its own it folds together a smaller feature budget with the loss of bigram features. To separate the two we also score the top-50 model against a matched unigram-only full reference, which is a clean feature-budget comparison (50 unigrams versus all unigrams) with no bigram confound: the retained fraction stays in the same band, 89.7 to 94.5% (Table 4). Either way a model that reads word order or longer context could recover signal these features miss, so we read the score as a statement about the linear-recoverable signal rather than the absolute lexicality of the task. The claim we stand behind is the comparative and relative one: across all five corpora the signal available to the standard linear pipeline is concentrated in a few high-association tokens. One might expect this to be an artifact of benchmarks whose labels track surface text; we test that directly on an implicit-hate corpus, which conveys hate without overt slurs, and find that the retained fraction stays inside the same band while the shortcut tokens shift from slurs to identity and topical terms (with some coded terms among them; Appendix G).

**The shortcut is not a linear-model artifact: it survives in a transformer.** The score above is computed for a TF-IDF logistic-regression pipeline, which leaves open the obvious objection that a contextual model would not lean on a handful of tokens. We test this directly on the same data. Holding the stratified 80/20 split and the per-dataset top-50  $\chi^2$  token set fixed (selected exactly as above, on the training split), we fine-tune `distilbert-base-uncased` twice: once on the full text and once on a *shortcut-only* copy of each post in which every word outside the 50-token set is deleted and the surviving tokens are kept in order. Only the visible vocabulary changes between the two runs; the model, split, labels, and training budget are identical (CPU, three epochs, two seeds). If the transformer needed word order or context beyond those tokens, the shortcut-only model would fall well short of the full-text one. It does not: on

Davidson, HateXplain, and TweetEval-Hate the shortcut-only distilBERT retains 92.4%, 90.4%, and 91.1% of its full-text macro-F1 respectively (full-text 0.93, 0.76, 0.76; per-seed retention 89.7 to 92.9%), in the same band as the linear shortcut score (90.4 to 92.4% for the transformer, versus 89.2 to 95.4% for the linear score). Extending the identical protocol (same split, token selection, and training budget) to the remaining two corpora closes the coverage to all five datasets (`transformer_shortcut_5ds.json`, which reproduces the three numbers above to within their two-seed confidence intervals): the shortcut-only distilBERT retains 88.8% on Stormfront and 81.4% on DynaHate. DynaHate is the lowest of the five, exactly as one would expect for the single corpus constructed adversarially to defeat surface cues, yet even there a contextual model seeing only fifty tokens per post keeps four-fifths of its full-text macro-F1. A 66M-parameter contextual model trained on only those fifty words per post thus recovers most of the performance it reaches on the full text across all five datasets, so the concentration of the recoverable signal is a property of the labeled benchmarks rather than an artifact of the bag-of-words pipeline (`transformer_shortcut.json` for the three slur-heavy corpora, `transformer_shortcut_5ds.json` for all five). To check that this is not merely an artifact of distilBERT’s small capacity, we re-ran the identical protocol (same stratified split, per-dataset top-50  $\chi^2$  token set, and three-epoch budget) with `roberta-base`, a 125M-parameter model roughly twice its size, on Apple-Silicon MPS and across three model-initialization seeds (`transformer_shortcut_robertabase.json`). The concentration survives the capacity increase and the per-dataset ordering is preserved: the shortcut-only `roberta-base` retains 92.2% (Davidson), 90.9% (HateXplain), 88.8% (TweetEval-Hate), 86.5% (Stormfront), and 78.3% (DynaHate) of its full-text macro-F1, again lowest on the adversarially debiased DynaHate. These are MPS-run numbers, reported as a larger-model robustness check rather than a CPU-reproducible result; the CPU distilBERT figures above remain the headline. We do not over-read this: both are base-size encoders fine-tuned for only three epochs, and the very large models trained longer that top modern leaderboards remain to be checked; what

| Dataset    | Full F1<br>(1,2) | Top-50 F1 | Ret. %<br>(1,2)   | Ret. %<br>(1,1) |
|------------|------------------|-----------|-------------------|-----------------|
| Davidson   | 0.899            | 0.816     | 90.8 [89.0, 92.5] | 90.5            |
| DynaHate   | 0.640            | 0.610     | 95.4 [93.1, 97.6] | 91.5            |
| HateXplain | 0.746            | 0.666     | 89.2 [86.8, 91.8] | 89.7            |
| TweetEval  | 0.734            | 0.692     | 94.4 [91.2, 97.6] | 94.5            |
| Stormfront | 0.681            | 0.645     | 94.7 [89.7, 99.9] | 93.8            |

Table 4: Lexical-shortcut reliance. A 50-token unigram model retains roughly 89 to 95% of full macro-F1 against the unigram-plus-bigram (1,2) reference used throughout (bootstrap 95% CI in brackets), and 89.7 to 94.5% against a matched unigram-only (1,1) reference. The unigram-only reference is a clean feature-budget comparison (50 unigrams vs. all unigrams) and removes the bigram confound; the retained fraction barely changes, and the apparent outlier (DynaHate, 95.4 vs. 91.5%) is the case where adding bigrams slightly *lowered* the full model.

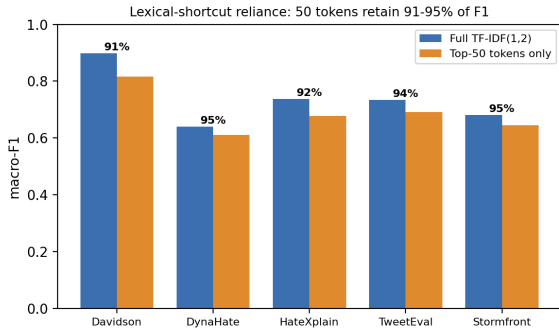


Figure 2: Full TF-IDF model vs. a 50-token model. Labels show retained percentage.

it removes is the specific worry that the shortcut is merely a TF-IDF limitation, and that it now holds even on the adversarially debiased DynaHate strengthens that reading.

#### 4.4 Transfer is poor, and not because of contamination

Figure 3 gives the transfer matrix. Mean in-dataset macro-F1 (held-out diagonal) is 0.738, while mean cross-dataset macro-F1 is 0.537, a gap of 0.20. Some pairs are far worse: a model trained on the Stormfront forum scores 0.29 on Davidson. Because cross-dataset duplication is negligible (Section 4.1), this gap is not contamination. It instead reflects two intertwined causes our design cannot fully separate: distributional shift, since the corpora differ in platform, sampling, topic, and time period, and definitional divergence, since the same texts receive conflicting labels (Section 4.2). The label-conflict measurement gives a lower bound on the definitional component, since it can only be

| Dataset    | Highest- $\chi^2$ tokens (slurs masked)                           |
|------------|-------------------------------------------------------------------|
| Davidson   | trash, b***h, monkey, h**s, oreo, colored                         |
| DynaHate   | hate, women, muslims, blacks, gays, foreigners                    |
| HateXplain | n***r, k**e, f****t, r*****d, queer, jew                          |
| TweetEval  | illegal, immigrant, deportthemall, illegals, buildthewall, deport |
| Stormfront | n***o, jews, ape, scum, filth, whites                             |

Table 5: Tokens most associated with the positive label. Slurs and swear words are masked with asterisks (first and last letter kept). The shortcut signal is slurs, identity terms, and a few topical keywords.

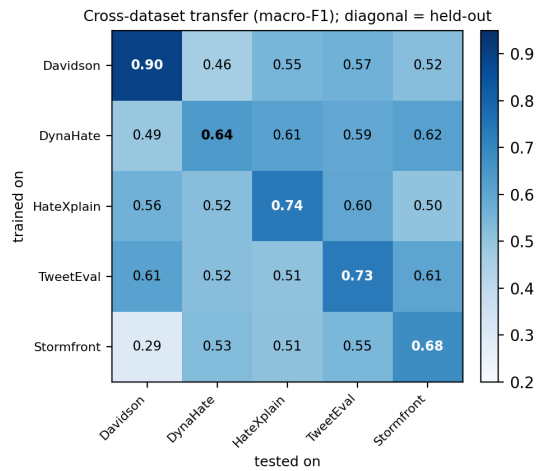


Figure 3: Cross-dataset transfer (macro-F1). Diagonal is a held-out in-dataset split. Off-diagonal transfer is much weaker.

evaluated on the small set of texts that happen to recur across corpora and need not be representative of the rest; we do not attempt to apportion the remaining gap precisely between definition and distribution. One reading to head off: the conflicting texts cannot mechanically account for the gap on their own, because the recurring texts on which conflict is even measurable are only 212 of 108,946 unique items (under 0.2%), too small a mass to move a 0.20 macro-F1 gap directly. We therefore read the 48.6% conflict rate not as the mechanism behind poor transfer but as a visible window onto a definitional divergence that, by construction, we can observe only where texts recur and that operates across the rest of the distribution we cannot probe directly. Either way, a model tuned to one corpus’s vocabulary and conventions does not carry over.



#### 4.5 Leakage is common but low-impact for linear models

Near-duplicate train/test leakage is real and sometimes substantial (9.8% of the TweetEval test set, 8.7% of DynaHate; full table in Appendix D), yet removing leaked items changes macro-F1 by at most 0.012, and the change is not consistently positive. For the standard linear TF-IDF pipeline we study, leakage does not meaningfully inflate scores (full per-dataset table in Appendix D). To check that this is not a quirk of the linear model, we repeated the leakage test with a fine-tuned distilBERT on the two official-split datasets; the inflation stays negligible ( $-0.0025$  on TweetEval-Hate and  $-0.0060$  on DynaHate, both within noise of zero across three seeds; Appendix H). So for a fine-tuned distilBERT on these two corpora the leakage inflation is also within noise of zero. We do not read this as settling the question for transformers in general: a 66M-parameter model trained for three epochs is about the least likely case to memorize its training set, so larger or longer-trained models could behave differently, and we still do not read it as license to skip deduplication.

### 5 Discussion: A Report Card for Abuse Datasets

Together the findings give a simple diagnosis for the five English benchmarks and pipelines we study (Table 6); we state it at that scope rather than as a claim about abuse benchmarks in general. The data-hygiene problems the community tends to police, cross-corpus contamination and train/test leakage, are either absent or, for the linear pipeline we study, inconsequential here. The problems that remain are conceptual. The first is model-independent: the datasets disagree about what abuse is, so a single-benchmark score conflates a corpus’s definition of the task with the quality of the model, and a model that tops one benchmark need not transfer to another. The transfer gap on its own does not carry this, because our corpora differ in platform, sampling, topic and time period as well as in definition, and this design cannot separate those causes. What carries it is the item-level comparison, where the surface text is held fixed, and that evidence is deliberately narrow: at the strict cutoff it rests on 18 shared items that reduce to six distinct texts (Table 7), and varying only the binary mapping moves the rate from 55.6% to 22.2% (Table 2). We therefore offer the conceptual reading as the one these

| Dataset    | Short.<br>ret.% | In<br>F1 | Out<br>F1 | $\Delta$ | Leak<br>% |
|------------|-----------------|----------|-----------|----------|-----------|
| Davidson   | 90.8            | 0.899    | 0.524     | 0.375    | 2.4       |
| DynaHate   | 95.4            | 0.640    | 0.579     | 0.061    | 8.7       |
| HateXplain | 89.2            | 0.746    | 0.541     | 0.205    | 0.4       |
| TweetEval  | 94.4            | 0.734    | 0.560     | 0.174    | 9.8       |
| Stormfront | 94.7            | 0.681    | 0.470     | 0.211    | 2.2       |

Table 6: Per-dataset report card. Short. ret.: top-50 shortcut retention; In F1: held-out in-dataset macro-F1; Out F1: mean macro-F1 transferring to the other four datasets;  $\Delta$ : transfer drop (In minus Out); Leak: train/test near-duplicate rate. Davidson is the most idiosyncratic (largest drop).

five benchmarks support, not as a demonstration that definitional disagreement is the largest single source of transfer loss. The second is scoped to our pipeline: for a standard linear baseline the score is largely keyword-driven, so the leaderboard number certifies mostly that the model has learned that corpus’s vocabulary, a concentration that persists for a fine-tuned distilBERT and a twice-as-large roberta-base (§4.3).

We package the audit as `abuse-audit`, a one-command tool that runs the near-duplicate, leakage, and shortcut checks on an arbitrary dataset CSV and emits a report card, with bundled scripts for the cross-dataset transfer check. It supports three lightweight practices: deduplicate splits and report the residual near-duplicate rate; report the shortcut score alongside accuracy as a one-number indicator of how lexical a benchmark is; and, when claiming generalization, report a cross-dataset transfer cell rather than only in-dataset performance. None requires new annotation or compute. Were a venue to mandate one, the near-duplicate rate is the candidate: it is a property of the split alone and cheap to compute, whereas the shortcut score and the transfer cell depend on a model choice and are better reported than required.

### 6 Conclusion

For the five English benchmarks and pipelines we audit, the measurement foundation is shakier than the data volume suggests: contamination and train/test leakage are minor, while the signal is concentrated in a few tokens and the datasets disagree on what abuse is. We do not claim this beyond that scope; other languages, gated or multimodal corpora, and the very large models that top current leaderboards are the extensions the `abuse-audit` tool supports.

## 7 Limitations

Our classifiers are primarily linear. To probe whether the leakage finding survives beyond linear models, we repeated the train/test leakage experiment with a fine-tuned distilBERT on the two official-split datasets (Appendix H); the macro-F1 inflation from leaked items remains negligible ( $-0.0025$  on TweetEval-Hate and  $-0.0060$  on DynaHate, both within noise of zero across three seeds). This extends the finding to one fine-tuned distilBERT, but does not settle it for transformers in general: a 66M-parameter model trained for three epochs is the case least likely to memorize, so it is the weakest possible test of the memorization worry the experiment is meant to address, and larger or longer-trained models remain to be checked. The shortcut result, linear by construction, now also holds for a fine-tuned distilBERT and, as a larger-model robustness check, for a roughly twice-as-large 125M roberta-base trained on shortcut-only text (§4.3); confirming it for the very large pretrained models trained longer than top modern leaderboards is the natural next step. We study five English datasets and a binary label mapping, which compresses fine-grained schemes and excludes multilingual and multimodal abuse; we add OLID (Zampieri et al., 2019) as a sixth-corpus robustness check above, and other widely used English benchmarks such as HatEval (Basile et al., 2019) could extend the pipeline further. Our five benchmarks tie their labels to surface text; implicit-hate corpora such as Latent Hatred (ElSherief et al., 2021), which deliberately target hate expressed without overt slurs, are the hardest case for any lexical-shortcut claim, so rather than leave them out of scope we ran the same shortcut pipeline on Latent Hatred (Appendix G). The absolute ceiling is lower there, as expected, but the retained fraction stays inside the band of the lexical benchmarks, with the shortcut tokens shifting from slurs to identity and topical terms (some of them coded), which sharpens rather than overturns the body’s claim. The shortcut score is one operationalization (top-50  $\chi^2$  unigrams against a unigram-plus-bigram reference); other choices would shift absolute numbers though we expect the qualitative pattern to hold, and we provide a threshold sensitivity analysis for the overlap measurements. Near-duplicate detection depends on a similarity threshold; we mitigate this by sweeping thresholds and verifying candidates with exact Jaccard. Finally, the bootstrap

confidence intervals throughout are marginal and descriptive: each is a per-quantity 95% interval and none is corrected for the number of datasets and metrics reported, so they should not be read as having joint (simultaneous) 95% coverage. Two analyses we did not run would sharpen the definition-versus-distribution question, and the toolkit supports both: a platform- and time-matched transfer design, in which training and test corpora are restricted to the same source and collection window so that the remaining gap is attributable to annotation guidelines, and a coded comparison of the shared-text conflicts in Appendix A against a matched sample of non-shared transfer errors, to show where definition dominates and where distribution does. We also did not test instruction-tuned large language models as classifiers, so whether their scores concentrate on the same tokens is open. Finally, nothing in the three audit checks is specific to abuse: the shortcut score applies unchanged to any text-classification benchmark whose labels might track surface vocabulary (sentiment, misinformation, toxicity), with the token budget and the reference model re-chosen per task.

## 8 Ethical Considerations

We use only existing public datasets, introduce no new annotations, and collect no user data. The corpora contain slurs and harmful content; we report high-association tokens and a small set of shared texts because they are central to the shortcut and conflict findings, but we mask slurs and swear words in every displayed token and example, place a content warning at the start of the paper, and quote no user handles beyond the datasets’ own placeholder, following the workshop’s reporting policy. Our audit is intended to improve evaluation of systems that mitigate online harm. We note a dual-use risk common to abuse research: descriptions of lexical shortcuts could inform evasion. The mitigations we recommend (deduplication, shortcut reporting, cross-dataset evaluation) are defensive, and the underlying weaknesses are already documented in the literature we cite.

### Reproducibility

All numbers are produced by the released code over the five public datasets, with fixed seeds. The toolkit downloads and normalizes the data, runs the audit, and regenerates every table and figure on a CPU in minutes; the optional transformer

leakage check in Appendix H is also CPU-only, but fine-tuning distilBERT on a CPU takes appreciably longer. The software environment is pinned in `requirements.txt` and SHA-256 checksums of the released results are listed in `CHECKSUMS.sha256`. The raw corpora are not redistributed: the audited datasets are the public abuse corpora named in the paper, fetched by the download script at the versions and access date (June 2026) recorded in the repository, which pins them, and the raw and normalized text is regenerated locally rather than shipped. To avoid republishing verbatim user content, the two intermediate artifacts that embed text are kept local and not released; the published results are derived statistics and report cards only. The code (including the `abuse-audit` tool) and the derived statistics and report cards are publicly available under a permissive license, with pointers to the original datasets under their respective terms; the code and derived artifacts are at <https://github.com/dhruvpate11706/abuse-audit>; an archived release with a citable DOI will be linked from that repository.

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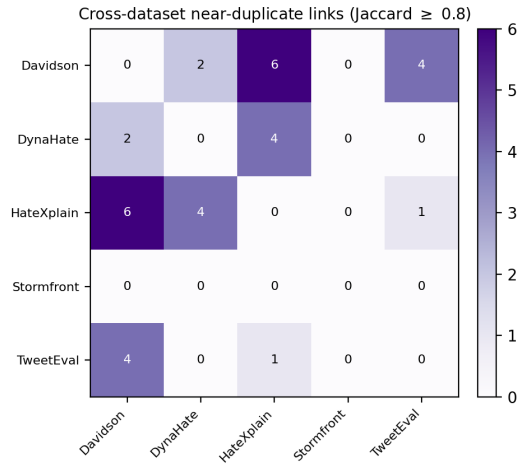


Figure 4: Cross-dataset near-duplicate links (Jaccard  $\geq 0.8$ ). Overlap is tiny and dominated by the Twitter-sourced corpora, though DynaHate also appears in 6 of the 17 pairs; Stormfront has none.

## A Every strict-cutoff label conflict, with a reading

Table 7 gives all nine conflicting cross-dataset near-duplicate links at  $J \geq 0.8$ , grouped by distinct links. Punctuation-only variants of the same sentence are collapsed into one row, which is why nine links become six rows. The table is generated by `code/make_conflict_examples.py`; only the italic reading lines are authorial. Slurs and swear words are masked with asterisks, following the workshop’s reporting policy; the underlying strings are the datasets’ normalized texts, and user handles appear only as the datasets’ own @user placeholder. Read together, none of the six is a clearly inconsistent labeling under a shared definition: four follow from Davidson’s broad *offensive* class, and the two that do not involve Davidson (the self-directed post and the reclaimed in-group banter) are borderline cases on which the corpora’s guidelines differ rather than annotation errors.

## B Cross-dataset overlap by corpus pair

Figure 4 breaks the 17 strict-cutoff cross-dataset near-duplicate links down by corpus pair, supporting the negligible-contamination finding of Section 4.1.

## C Per-pair conflict severity

Table 8 breaks the strict-cutoff (Jaccard  $\geq 0.8$ ) cross-dataset label conflicts down by dataset pair, supporting Section 4.2.



| Shared text (normalized)                                                                                                                                                                        | Abuse in   | Not abuse in         | Links | $J$  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|-------|------|
| karma is a b***h<br><i>Generic profanity with no target. Davidson’s broad offensive class marks it positive; the hate-specific corpora do not.</i>                                              | Davidson   | DynaHate, HateXplain | 4     | 1.00 |
| i f*****g hate myself<br><i>Self-directed, no target group. Not an offensive-versus-hate split at all: one corpus reads the hate verb as the signal.</i>                                        | HateXplain | DynaHate             | 1     | 1.00 |
| @user you p***y<br><i>Gendered insult aimed at an individual, not a protected group.</i>                                                                                                        | Davidson   | TweetEval            | 1     | 0.92 |
| u a stupid h*<br><i>Gendered insult aimed at an individual. Same split as the line above.</i>                                                                                                   | Davidson   | TweetEval            | 1     | 0.90 |
| hell yea b***h dis go hard as hell flocka i will suck<br>big d**k team s**t all...<br><i>In-group AAVE-inflected banter. Reclaimed usage that the two corpora score in opposite directions.</i> | HateXplain | TweetEval            | 1     | 0.88 |
| she’s a h*e, she’s ugly, she’s a b***h, she’s this, she’s that .. but you...<br><i>Quoted misogynistic speech that the post itself is criticising.</i>                                          | Davidson   | TweetEval            | 1     | 0.84 |

Table 7: Every cross-dataset near-duplicate link at the strict cutoff ( $J \geq 0.8$ ) whose endpoints disagree on the binary label. The 9 conflicting links reduce to 6 distinct texts, and one text (“karma is a b\*\*\*h”) accounts for 4 of them. Italic lines are our reading of each case. Most are the broad offensive-versus-hate split, but two are not: one is self-directed and one is reclaimed in-group usage. Slurs and swear words are masked with asterisks following the workshop’s reporting policy. Generated by `code/make_conflict_examples.py`.

|            | Dav | Dyn | HtX | Stf | TwE |
|------------|-----|-----|-----|-----|-----|
| Davidson   | 0   | 2   | 2   | 0   | 3   |
| DynaHate   | 2   | 0   | 1   | 0   | 0   |
| HateXplain | 2   | 1   | 0   | 0   | 1   |
| Stormfront | 0   | 0   | 0   | 0   | 0   |
| TweetEval  | 3   | 0   | 1   | 0   | 0   |

Table 8: Per-pair conflict severity: number of cross-dataset near-duplicate links (Jaccard  $\geq 0.8$ ) whose two endpoints carry different binary labels, for each dataset pair. The matrix is symmetric; columns abbreviate the five datasets. All 9 conflicting links involve Davidson, DynaHate, HateXplain, or TweetEval; Stormfront has no cross-dataset near-duplicate at this threshold and so contributes no conflicts. Davidson, whose broad “offensive” class is the main source of disagreement, participates in 7 of the 9.

## D Train/test leakage table

Table 9 gives the per-dataset train/test near-duplicate leakage rate and its effect on macro-F1 for the linear pipeline, summarized in Section 4.5. Leakage is real and sometimes substantial (up to 9.8% of a test set) yet removing leaked test items shifts macro-F1 by at most 0.012 in magnitude.

## E Near-duplicate threshold sensitivity

Table 10 reports duplicate-pair counts at four Jaccard thresholds. Cross-dataset pairs stay in the low tens at every threshold, while within-dataset pairs are two to three orders of magnitude larger, confirming that the negligible-contamination finding is not an artifact of a particular cutoff. The final col-

| Dataset    | Split    | Leaked % | F1    | F1 dedup |
|------------|----------|----------|-------|----------|
| Davidson   | random   | 2.4      | 0.897 | 0.896    |
| DynaHate   | official | 8.7      | 0.575 | 0.586    |
| HateXplain | random   | 0.4      | 0.733 | 0.733    |
| TweetEval  | official | 9.8      | 0.401 | 0.400    |
| Stormfront | random   | 2.2      | 0.684 | 0.682    |

Table 9: Train/test near-duplicate leakage and its effect. Removing leaked test items shifts macro-F1 by at most 0.012 in magnitude (−0.0113 on DynaHate).

umn gives the pair-weighted cross-dataset conflict count for reference; the item-level conflict rates we lead with in Section 4.2 are computed over distinct nodes rather than pairs.

| Jaccard | All pairs | Cross | Within | Cross conflict |
|---------|-----------|-------|--------|----------------|
| 0.5     | 43,843    | 272   | 43,571 | 132            |
| 0.7     | 15,805    | 26    | 15,779 | 11             |
| 0.8     | 5,218     | 17    | 5,201  | 9              |
| 0.9     | 1,097     | 12    | 1,085  | 5              |

Table 10: Duplicate pair counts vs. Jaccard threshold.

## F Shortcut budget sensitivity and majority baseline

Table 11 reports retained macro-F1 (top- $k$  model vs. full model) as the token budget  $k$  varies, alongside a majority-class baseline. On the four larger datasets retention rises monotonically with  $k$  and is already high at  $k = 10$ ; the small Stormfront set (2,133 test items) is nearly flat and slightly non-monotone (it dips from 93.4% at  $k = 10$  to 92.8% at  $k = 25$ ),

which we read as sampling noise rather than a real trend. In every case the 50-token operating point used in the body is representative rather than cherry-picked. The majority-class macro-F1 (0.35 to 0.47) is far below even the 10-token model, confirming that the retained signal is genuine rather than an artifact of class imbalance.

| Dataset    | Maj. F1 | Full F1 | @10  | @25  | @50  | @100 |
|------------|---------|---------|------|------|------|------|
| Davidson   | 0.45    | 0.899   | 62.7 | 86.9 | 90.8 | 92.9 |
| DynaHate   | 0.35    | 0.640   | 85.4 | 92.6 | 95.4 | 99.8 |
| HateXplain | 0.38    | 0.746   | 76.0 | 85.6 | 89.2 | 92.1 |
| TweetEval  | 0.37    | 0.734   | 88.9 | 92.5 | 94.4 | 95.3 |
| Stormfront | 0.47    | 0.681   | 93.4 | 92.8 | 94.7 | 94.6 |

Table 11: Shortcut retention (% of full macro-F1) at token budgets  $k \in \{10, 25, 50, 100\}$ , with a majority-class macro-F1 baseline. Computed on the single held-out split (seed 42).

## G Implicit hate: does the shortcut survive?

The five audited benchmarks tie their labels to surface text, so it is fair to ask whether the lexical-shortcut finding is an artifact of that design. The natural stress test is implicit hate, which deliberately conveys hate without overt slurs. We therefore ran the identical shortcut pipeline (stratified 80/20, seed 42, full TF-IDF unigram-plus-bigram reference vs. a 50-token  $\chi^2$ -unigram model, macro-F1) on the stage-1 binary file of the Latent Hatred corpus (ElSherief et al., 2021), obtained from a public, ungated mirror on the Hugging Face Hub.<sup>1</sup> Our primary task classifies *implicit\_hate* (positive) against *not\_hate* (negative), excluding the lexical *explicit\_hate* class so the test isolates hate expressed without overt slurs; we also report an any-hate framing (implicit and explicit together vs. *not\_hate*) for completeness.

The result is informative, and not the one a strong version of the “it is all lexical” reading would predict. As Table 12 shows, the implicit task is genuinely harder in absolute terms: full-model macro-F1 is 0.712, below four of the five lexical benchmarks, and it clears the majority-class baseline (0.395) by only 0.317. Yet the *retained fraction* does not collapse: 50 tokens still recover 91.5% of full-model F1 (bootstrap 95% CI [88.7, 94.2]),

<sup>1</sup>[tasksources/implicit-hate-stg1](https://tasksources/implicit-hate-stg1), a redistribution of the original SALT-NLP/implicit-hate stage-1 release; 21,480 posts labeled *not\_hate* (13,291), *implicit\_hate* (7,100), or *explicit\_hate* (1,089).

squarely inside the 89.2 to 95.4% band of the lexical benchmarks. The any-hate framing behaves the same way (0.720 full, 93.6% retained). What changes is the *content* of the shortcut rather than its strength: the highest- $\chi^2$  tokens, in order, are *illegals*, *them*, *jews*, *illegal*, *deport*, *jew*, *whitegenocide*, *diversity*, *black*, *genocide*, *blacks*, *send*, *whites*, *muslims*, *civilization*. These are mostly plain identity and topical terms, with some coded or dog-whistle terms among them (e.g. *whitegenocide*, *diversity*), rather than the raw slurs that top the lexical benchmarks. We read this as a sharpening, not a refutation, of the body’s claim: once an implicit-hate corpus is annotated, its labels still leave a recoverable lexical fingerprint for a linear TF-IDF model, only in a different vocabulary, so the shortcut score is not escaped simply by collecting examples that lack overt slurs. The lower absolute ceiling is where implicitness shows up, and a non-linear encoder that reads context is where we would expect the remaining gap to lie, consistent with the linear-model scope we keep throughout.

| Task                  | Maj. F1 | Full F1 | Retained %        |
|-----------------------|---------|---------|-------------------|
| Implicit vs. not-hate | 0.395   | 0.712   | 91.5 [88.7, 94.2] |
| Any-hate vs. not-hate | 0.382   | 0.720   | 93.6 [90.9, 96.4] |

Table 12: Lexical-shortcut check on the Latent Hatred implicit-hate corpus (ElSherief et al., 2021), computed identically to Table 4 (50-token  $\chi^2$  model vs. full TF-IDF; bootstrap 95% CI on the retained fraction in brackets). Absolute F1 is lower than on the lexical benchmarks, but the retained fraction stays inside their 89.2 to 95.4% band.

## H Transformer leakage check

To verify that the low-impact leakage finding (Table 9) is not an artifact of the linear pipeline, we re-ran the train/test near-duplicate leakage experiment with a fine-tuned *distilbert-base-uncased* (three epochs, three seeds) on the two official-split datasets, TweetEval-Hate and DynaHate. As Table 13 shows, removing leaked items does not reduce macro-F1: the inflation is slightly negative on both (−0.0025 and −0.0060), within noise of zero. This is one small, lightly fine-tuned transformer, the case least prone to memorization, so it extends the linear finding rather than settling it for larger models.

| Dataset        | $n_{\text{test}}$ | Leaked % | Full F1 | Dedup F1 | Infl.   |
|----------------|-------------------|----------|---------|----------|---------|
| TweetEval-Hate | 2,970             | 9.8      | 0.453   | 0.456    | -0.0025 |
| DynaHate       | 4,120             | 8.7      | 0.759   | 0.765    | -0.0060 |

Table 13: Transformer leakage check. Fine-tuned `distilbert-base-uncased`, mean of three seeds: removing train/test near-duplicate leakage does not inflate macro-F1, matching the linear result in Table 9. Inflation is full minus dedup; the across-seed standard deviation of the inflation is at most 0.0013 on both datasets.